Lila Aryadwita

CHE 210 - HW 2

① 1.12 Benzene-to-adipic → \$ 27,100 / day
Glucose-to-adipic → - \$ 5,400 / day

- $C_6H_{12}O_6$ → - \$ 48,500 / day
  Glucose process to be competitive = 27,100 + 5,400
  = \$ 32,500

  48500 - 32,500 = \$ 16,000 kg/day

  $\$16{,}000 \frac{kg}{day} \times \frac{1 \text{ day}}{80{,}850}$

  = $\boxed{\$ 0.198 / kg}$ ✓

- Glucose - to- catechol → \$ 49,200
  Benzene-to-catechol → \$ 89,300

  89,300 - 49,200 = \$ 40,100
  \$ 48,500 - \$ 40,100 = \$ 8,400 kg/day

  $\$ 8{,}400 \frac{kg}{day} \times \frac{1 \text{ day}}{80{,}770 \text{ kg}} = \boxed{\$ 0.104 / kg}$ ✓

② 1.34 gasoline = 25 million tons/year
high fructose corn syrup = 15 million tons/year
aspirin = 25000 tons/year

-3 . Need provide a logic.

- I think plastic has the highest US production rates because plastic is used in many industries and households and has many uses.

→ $\boxed{\text{organic} = 6\times10^{10} \text{ tons/year}}$ BIGGEST ↑ plastics = 34.5 m tons/year, Synthetic fibers = 900B tons/yr, inorganic = 5.8 m tons/year ✓

Actual:

- gasoline = 16.7 million tons/year (Chemical and Engineering News)
- HFCS = 8.3 million tons/year (USDA)
- Aspirin = 40,000 tons/year

According to table 1.4, the typical plant capacities for petroleum and pharmaceuticals are bigger than the market size so the industry is able to fulfill the market's needs.

(3) 2.11 1970 → 14 g/mile (1100 billion miles)
2000 → 1 g/mile (2600 billion miles)
total hydrocarbon emission in 1970 & 2000? (tons)
% increase/decrease over 30 yr period?

• 1970 $\frac{14\ g}{1\ mile} \times 1100 \times 10^9\ mile \times \frac{1.102 \times 10^{-6}\ ton}{1\ g} = \boxed{16970800\ tons}$ ✓

• 2000 $\frac{1\ g}{mile} \times 2600 \times 10^9\ mile \times \frac{1.102 \times 10^{-6}\ ton}{1\ g} = \boxed{2865200\ tons}$ ✓

• % increase/decrease $= \frac{2865200 - 16970800}{16970800} \times 100\%$

$= \boxed{83\% \text{ DECREASE } (-83\%)}$ ✓

(4) $\ln \frac{C_A - C_{A\infty}}{C_{A0} - C_{A\infty}} = -Kt$ (proposed reaction mechanism)

a) $\ln\left(\frac{0.1025 - 0.0495}{0.1823 - 0.0495}\right) = -K(100)$ $\boxed{K \rightarrow 9.2 \times 10^{-3}\ g/l\,min}$

$\ln\left(\frac{C_{A65} - 0.0495}{0.1823 - 0.0495}\right) = -(9.2 \times 10^{-3})(65)$

$\frac{C_{A65} - 0.0495}{0.1823 - 0.0495} = 0.5499$ /6

$C_{A65} = 0.1225\ g/l$

→ The data supports the prediction using the proposed reaction mechanism because the concentration at t = 65 min using the formula is 0.1225 g/l ≈ 0.1216 g/l

b) A → B, tank volume 30.5 gal, no B at t = 0, 2 hours?

$30.5\ gal \times \frac{3.785\ l}{1\ g} = 115.4425\ l$

$\ln\left(\frac{C_{A120} - 0.0495}{0.1823 - 0.0495}\right) = -(9.2 \times 10^{-3})(120)$

$\frac{C_{A120} - 0.0495}{0.1823 - 0.0495} = 0.332 \rightarrow C_{A120} = 0.094\ g/l$ ✓

76/85

- $C_B = C_A - C_{A120}$
  $= 0.1823 - 0.094$
  $= 0.0883 \ g/l$
- $0.0883 \frac{g}{l} \times 115.4425 \ l = \boxed{10.19 \ g}$ at B(120)

(5) mixture of ethanol and water contains 60 wt% water
a) specific gravity of the mixture at 20°C ?
vol (l) of mixture is required to provide 150 moles ethanol?

- $SG = \frac{\rho_{mix}}{\rho_{ref}}$ $\rho_{ref} = 1 \ g/cm^3$ for H_2O @ 4°C

Water: $\frac{60}{100} = 60\% \ H_2O$

Ethanol: $\frac{40}{100} = 40\% \ C_2H_6O$

$\rho_{ethanol} = 0.79 \ g/cm^3$
$\rho_{H_2O} = 0.997 \ g/cm^3$

- $V_{mix} = V_{ethanol} + V_{water}$
  $= \left(40g \times \frac{1 cm^3}{0.79g}\right) + \left(60g \times \frac{1 cm^3}{0.997g}\right)$

$V_{mix} = 110.8 \ cm^3$

- $\rho_{mix} = \frac{M_{mix}}{V_{mix}} = \frac{100g}{110.8 cm^3} = 0.903 \ g/cm^3$
- $SG = \frac{0.903 g/\cancel{cm^3}}{0.997 \ \cancel{g/cm^3}} = \boxed{0.906 = SG}$

- $\frac{150 \ \cancel{mol}}{C_2H_6\cancel{O}} \times \frac{46.6 \ \cancel{g \ ethanol}}{1 \ mol \ C_2H_6O} \times \frac{100 \ \cancel{g \ mixture}}{40 g \ ethanol} \times \frac{1 \ \cancel{cm^3}}{0.903 \ \cancel{g \ mix}} \times \frac{1 \ l}{1000 \ \cancel{cm^3}}$
  $= \boxed{19.35 \ l \ \text{mixture}}$

- $\frac{150 \ mol}{C_2H_6O} \times \frac{46.6 \ g \ ethanol}{1 \ mol \ C_2H_6O} \times \frac{100 g \ mixture}{40 g \ ethanol}$
  $= 17475 \ g$ mixture

$$0.93518 = \frac{\rho_{mix}}{1\ g/cm^3}$$

$\rho_{mix} = 0.93518\ g/cm^3$

b) SG mixture at 20°C = 0.93518

% error?

$$\%\ error = \frac{exp - theo}{theo} \times 100$$

$$17475g \times \frac{1\ cm^3}{0.93518g} \times \frac{1\ l}{1000\ cm^3} = 18.7\ l$$

$$\%\ error = \left(\frac{19.35 - 18.7}{18.7}\right) \times 100$$

$$= \boxed{3.5\%\ error}$$

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